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Pathophysiology and Etiology of Acute Kidney Injury

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DEFINITION

Acute kidney injury (AKI) is defined as an abrupt decline in glomerular filtration rate (GFR) sufficient to decrease the elimination of nitrogenous waste products and other uremic toxins. This has traditionally been referred to as *acute renal failure* (ARF) but has been replaced with the term *acute kidney injury*, and standardized definitions of AKI have been developed (see Table 72.1) with stages based upon the magnitude of the rise in serum creatinine and changes in the urine output over 1 week.[1]

ETIOLOGY

AKI is caused by a broad range of etiologies, and the differential diagnosis must be considered in a systematic fashion. The traditional paradigm divides AKI into prerenal, renal, and postrenal causes. Prerenal causes may be due to hypovolemia or a decreased effective arterial volume. Postrenal kidney failure is usually due to obstruction. Intrinsic renal causes of AKI should be considered under the different anatomic components of the kidney (vascular supply, glomerular, tubular, and interstitial disease; Fig. 70.1). Major extrarenal artery or venous occlusion must also be considered (see Chapter 43). Disorders of the small intrarenal vasculature can also result in AKI (e.g., vasculitis, thrombotic microangiopathy, malignant hypertension, eclampsia, postpartum states, disseminated intravascular coagulation [DIC], scleroderma; see Chapters 26, 30, 38, 42, and 65). All forms of acute glomerulonephritis can present as AKI, as can acute inflammation and space-occupying processes of the kidney interstitium (e.g., drug-induced, infectious, and autoimmune disorders, leukemia, lymphoma, sarcoidosis).

Among inpatients, prerenal azotemia and acute tubular injury (ATI) account for most AKI cases[2] (Fig. 70.2), often in the setting of AKI superimposed on chronic kidney disease (CKD), so-called acute-on-chronic kidney disease. The term ATN (acute tubular necrosis), although commonly used, is a misnomer because the alterations are not limited to the tubular structures, and true cellular necrosis in human ATN is often minimal. ATN should be reserved for cases of AKI in which a kidney biopsy (if performed) shows the characteristic changes of tubular cell injury (Fig. 70.3), or for patients with findings of tubular injury (e.g., renal tubular epithelial cells in the urine sediment) in an appropriate clinical setting. There are also geographic differences in the causes of AKI, with the spectrum of causes in tropical countries described in Chapter 71.

PATHOPHYSIOLOGY AND ETIOLOGY OF PRERENAL AKI

Impaired kidney perfusion with a resultant fall in glomerular capillary filtration pressure is a common cause of AKI. In this setting, tubular function is typically normal, whereas kidney reabsorption of sodium and water is increased, so the urine exhibits low sodium concentration (<20 mmol/L) and high osmolality (>500 mOsm/kg), presuming a diuretic has not been administered. A marked reduction in kidney perfusion may overwhelm autoregulation and precipitate an acute fall in GFR. With lesser degrees of kidney hypoperfusion, glomerular filtration pressures and GFR are maintained by afferent arteriolar vasodilation (mediated by vasodilatory eicosanoids) and efferent arteriolar vasoconstriction (mediated by angiotensin II) that increases the filtration fraction. In this setting, AKI may be precipitated by agents that impair afferent arteriolar dilation (nonsteroidal antiinflammatory drugs [NSAIDs]) or efferent vasoconstriction (angiotensin-converting enzyme [ACE] inhibitors and angiotensin receptor blockers [ARBs]).

The hemodynamic mechanisms that drive the reduction in GFR and an adaptive increase in proximal tubular reabsorption in hypovolemic states originate from an interplay between hydraulic and oncotic pressures (Fig. 70.4). Under normal physiologic conditions, the glomerular capillary hydraulic pressure (P_{GC}) is greater than the glomerular capillary oncotic pressure (ϖ_{GC}), and hydraulic pressure in Bowman's space that opposes filtration and the oncotic pressure in Bowman's space are negligible. Thus, the sum of these forces leads to the generation of ultrafiltrate (see Chapter 2). The directionality of fluid flow reverses in the proximal tubule, where the peritubular capillary oncotic pressure (ϖ_{TC}) is greater than the peritubular capillary hydraulic pressure (P_{TC}), thus favoring reabsorption. As kidney perfusion falls beyond compensatory limits, P_{GC} falls. When volume loss is pronounced, ϖ_{GC} may rise as a result of hemoconcentration and rise in serum total protein concentration. Those processes lead to a fall in GFR. Concomitantly, the dominance of ϖ_{TC} over P_{TC} becomes more pronounced primarily as a result of further drop in P_{TC}. This avid proximal tubular reabsorption in prerenal azotemia partly explains the increased reabsorption of molecules like sodium, urea, and calcium observed in this disease state.

Prerenal AKI is often secondary to gastrointestinal losses (diarrhea, vomiting, prolonged nasogastric drainage), renal losses (diuretics, osmotic diuresis in hyperglycemia), dermal losses (burns, extensive sweating), or sequestration of fluid, sometimes known as third spacing (e.g., acute pancreatitis, muscle trauma). Kidney perfusion may also be impaired in the setting of normal or increased extracellular fluid, when

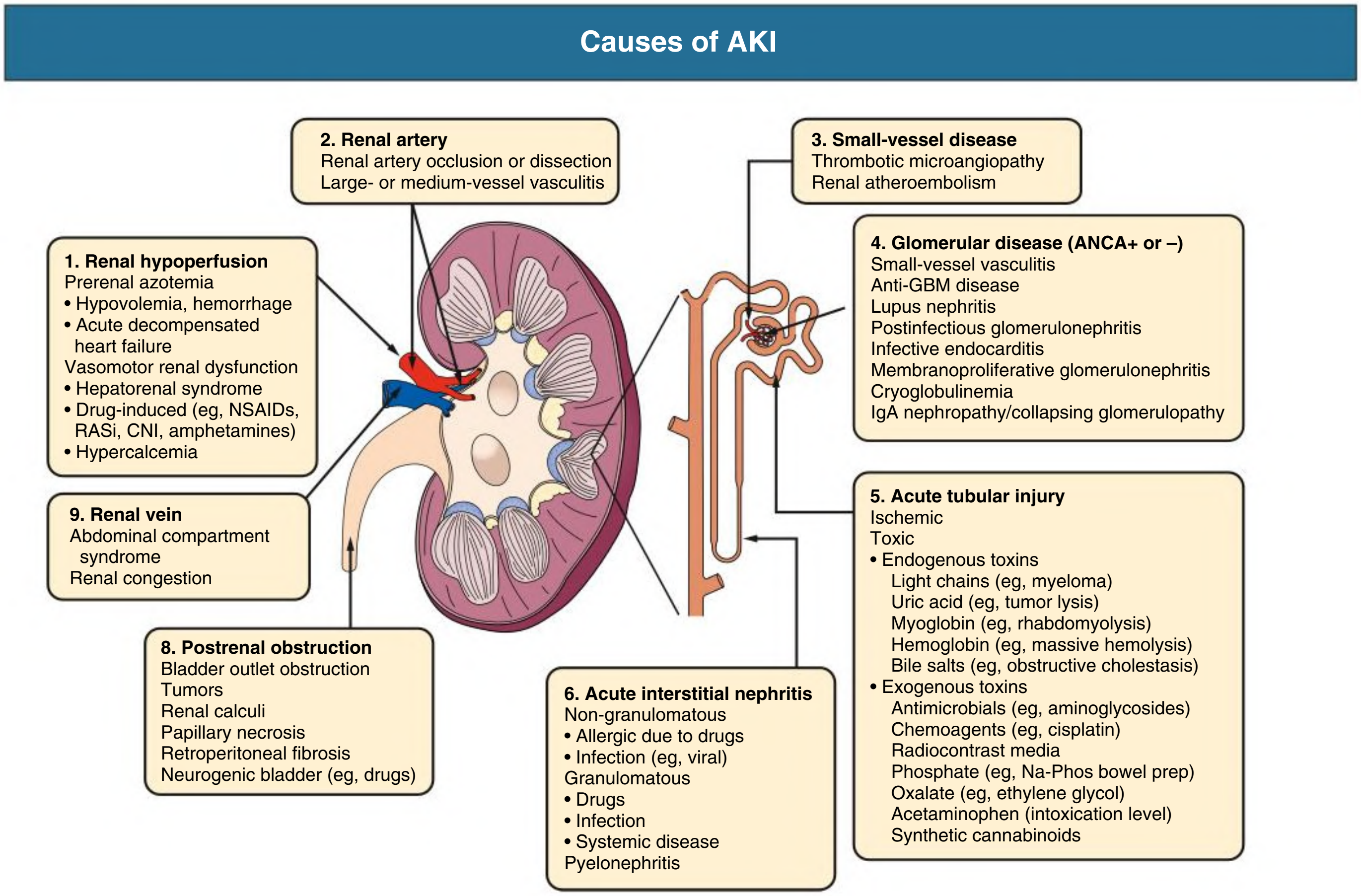

Fig. 70.1 Causes of Acute Kidney Injury (AKI). AKI is classified into prerenal, renal, and postrenal causes. Renal causes of AKI should be considered under the different anatomic components of the kidney (vascular supply, glomerular, tubular, and interstitial disease). *ANCA,* Antineutrophil cytoplasmic antibody; *CNI,* calcineurin inhibitor; *GBM,* glomerular basement membrane; *IgA,* immunoglobulin A; *Na-Phos,* sodium-phosphate; *NSAID,* nonsteroidal antiinflammatory drug; *RASi,* renal-angiotensin system inhibitor.

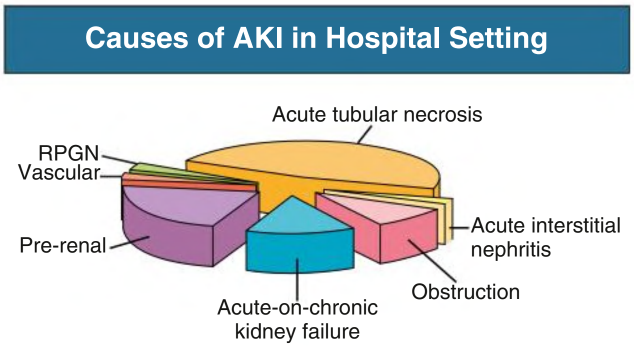

Fig. 70.2 Causes of acute kidney injury *(AKI)* in the hospital setting. Madrid acute renal failure study (1996). *RPGN,* Rapidly progressive glomerulonephritis.

cardiac output is reduced (heart failure), or when there is systemic arterial vasodilation with redistribution of cardiac output to extrarenal vascular beds (e.g., sepsis, liver cirrhosis) (see Chapters 8 and 76). An unusual cause of prerenal AKI is the infusion of large quantities of osmotically active substances such as mannitol, dextran, or protein, which can increase the glomerular oncotic pressure enough to exceed the capillary hydrostatic pressure stopping filtration.

Prerenal AKI can be corrected if the extrarenal factors causing the kidney hypoperfusion are rapidly reversed. Failure to restore renal blood flow (RBF) during the functional prerenal stage will ultimately lead to tubular cell injury.

PATHOPHYSIOLOGY OF ATI

ATI commonly occurs in patients with trauma, vascular and cardiac surgery, severe burns, pancreatitis, sepsis, and chronic liver disease. ATI is responsible for most cases of hospital-acquired AKI and is usually due to ischemic or nephrotoxic injury, or a combination of both. In the intensive care unit, two-thirds of cases of AKI are due to the combination of impaired kidney perfusion, sepsis, and nephrotoxic agents. The importance of combined injury is highlighted by animal studies, in which severe and prolonged hypotension or a single nephrotoxic agent may not cause ATI. Fever may exacerbate ATI by increasing the renal tubular metabolic rate, thereby increasing adenosine triphosphate (ATP) consumption. The typical course of uncomplicated ATI is recovery over 2 to 3 weeks; however, superimposed kidney insults (e.g., episodes of hypotension induced by hemodialysis) or multiple comorbidities often alter this pattern.

Histology

The typical features of ATI include vacuolization and loss of brush border in proximal tubular cells. Sloughing of tubular cells into the lumen leads to cast obstruction, manifested by tubular dilation. Interstitial edema can produce widely spaced tubules, and a mild leukocyte infiltration may be present (see Fig. 70.3). Despite the term acute tubular

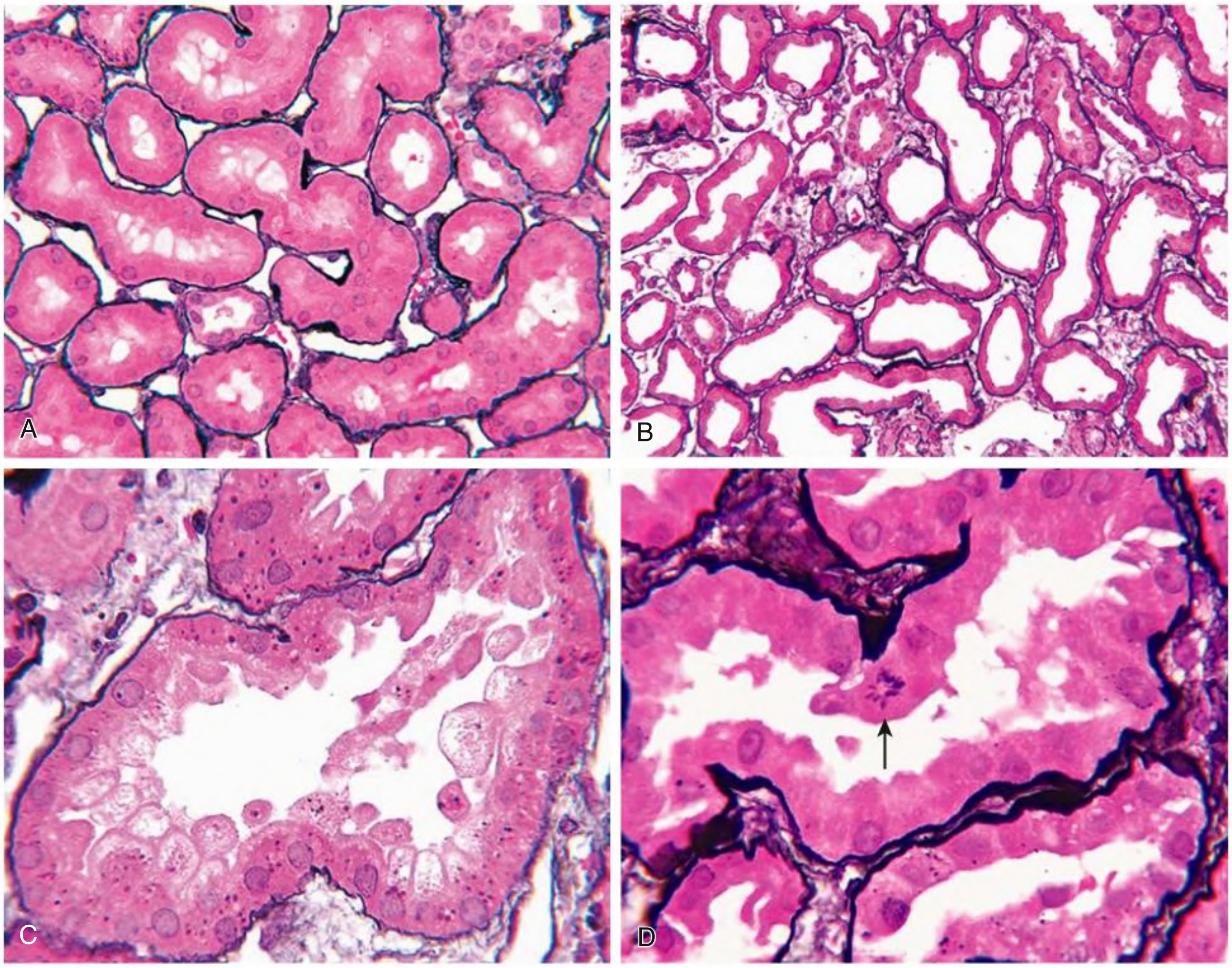

Fig. 70.3 Kidney Pathology in Acute Tubular Necrosis (ATN). (A) Normal cortical renal tubules. (B–C) ATN. Note the flattened epithelium, bare basement membranes, and intraluminal cellular debris. (D) Recovering ATN showing a tubular epithelial cell mitotic figure *(arrow)*. (Courtesy Erika Bracamonte, MD, University of Arizona.)

"necrosis," frankly necrotic cells are uncommon on kidney biopsy. Histologic evidence of injury frequently only involves 10% to 15% of the tubules despite marked functional impairment, implying that additional factors such as vasoconstriction and tubular obstruction are important in the loss of GFR.

Site of Tubular Damage in ATI

Tubular damage is usually due to a combination of ischemic injury resulting in depletion of cellular ATP and direct tubular epithelial cell injury by nephrotoxins. The S3 segment of the proximal tubule and the medullary thick ascending limb (mTAL) are particularly vulnerable to hypoxic injury (Fig. 70.5) for several reasons:

1. *Blood supply:* Most blood flow to the kidney is directed to the kidney cortex for glomerular filtration, where tissue Po_2 is 50 to 100 mm Hg. By contrast, the outer medulla and medullary rays are watershed areas receiving their blood supply from vasa rectae. Countercurrent oxygen exchange occurs, leading to a progressive fall in Po_2 from cortex to medulla. This results in medullary cells living on the "brink of hypoxia" (Po_2 as low as 10–15 mm Hg). The S3 segments of proximal tubule cells and distal mTALs are thus exposed to chronic borderline oxygen deprivation.
2. *High tubular energy requirements:* The cells of the S3 region and mTAL have high metabolic activity due to sodium reabsorption driven by basolateral membrane Na^+,K^+-ATPase. Blocking sodium reabsorption in the mTAL with loop diuretics raises the medullary tissue Po_2 from about 15 to 35 mm Hg. The low GFR in AKI may be renoprotective by diminishing sodium filtration and hence limiting the need for ATP-dependent sodium reabsorption. The drop in GFR in this setting has been called acute renal success. However, there is no clinical evidence that loop diuretics foster AKI recovery.
3. *Glycolytic ability of tubular cells:* Proximal tubular cells have minimal glycolytic machinery and rely almost solely on oxidative phosphorylation for the generation of ATP. In contrast, mTAL cells have a large glycolytic capacity and are more resistant to hypoxic or ischemic insults.

Hemodynamic Factors in the Development of ATI

Impaired Kidney Autoregulation

Autoregulation between systolic blood pressures of 80 and 150 mm Hg allows maintenance of RBF, glomerular pressures, and GFR in a stable range. Below 80 mm Hg, this autoregulation fails, and ischemic injury may result. In certain conditions, such as aging or CKD, autoregulation is impaired, and ischemic injury may occur more easily with reductions in perfusion pressure. In settings of low kidney perfusion (e.g., volume depletion, left ventricular failure, renal artery stenosis, chronic liver disease), GFR may be dependent on autoregulation mediated by vasodilatory prostaglandins acting on the afferent arteriole and angiotensin II–mediated efferent arteriolar vasoconstriction to maintain glomerular pressure. Any interference with these mechanisms (e.g., administration of ACE inhibitors or NSAIDs) may produce a precipitous fall in GFR.

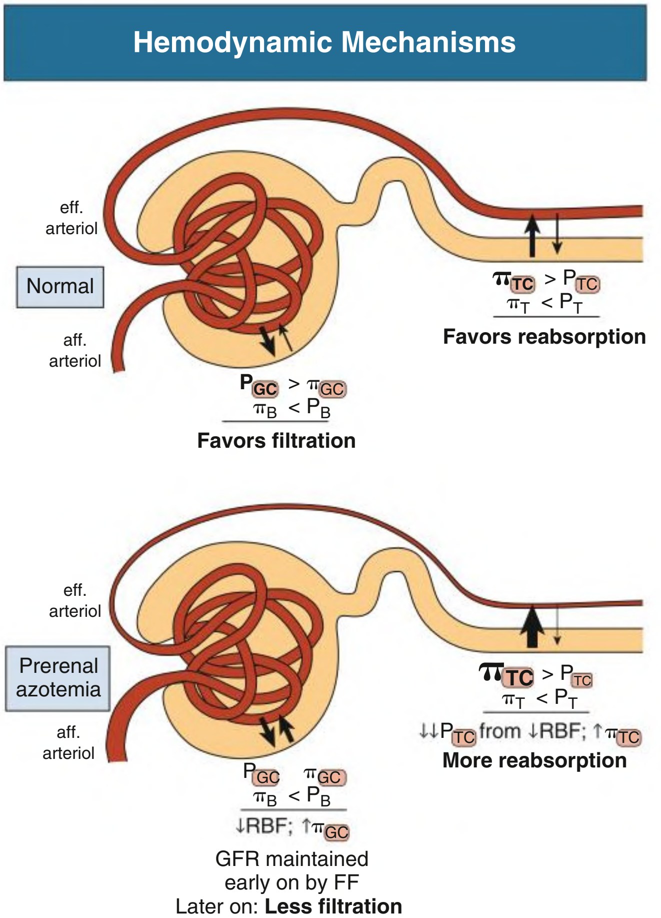

Fig. 70.4 Hemodynamic Mechanisms That Drive the Reduction in Glomerular Filtration Rate (GFR) and an Adaptive Increase in Proximal Tubular Reabsorption in Hypovolemic States Originate From an Interplay Between Hydraulic and Oncotic Pressures. Under normal physiologic conditions, the glomerular capillary hydraulic pressure (P_{GC}) that drives fluid from the capillary tuft into Bowman's space is greater than the glomerular capillary oncotic pressure (π_{GC}) that drives fluid into the glomerular capillary. The hydraulic pressure in Bowman's space (P_B) that opposes filtration and the oncotic pressure in Bowman's space (π_B) are of lesser magnitude or negligible. Thus, the sum of these forces leads to the generation of ultrafiltrate. The directionality of filtration fraction (FF) reverses in the proximal tubule where the peritubular capillary oncotic pressure (π_{TC}) is greater than the peritubular capillary hydraulic pressure (P_{TC}), thus favoring reabsorption. The hydraulic pressure inside the tubular lumen *(P_T)* and the oncotic pressure inside the tubular lumen (π_T) are of lesser magnitude. As kidney perfusion falls beyond compensatory limits, P_{GC} falls. When volume loss is pronounced, π_{GC} may rise as a result of hemoconcentration and rise in serum total protein concentration. Those processes lead to a fall in GFR. Concomitantly, the dominance of π_{TC} over P_{TC} becomes more pronounced primarily as a result of further drop in P_{TC}. This avid proximal tubular reabsorption in prerenal azotemia partly explains the increased reabsorption of molecules like sodium, urea, and calcium observed in this disease state. *RBF,* Renal blood flow.

Intrarenal Vasoconstriction

In established ATI, RBF is decreased by 30% to 50%. Indeed, in AKI, rather than the normal autoregulatory renal vasodilation that occurs in response to decreased perfusion pressure, there is evidence of renal vasoconstriction. Vasoconstrictors implicated in this response include angiotensin II, endothelin-1, adenosine, thromboxane A_2, prostaglandin H_2, leukotrienes C_4 and D_4, sympathetic nerve stimulation (Fig. 70.6), and tubuloglomerular feedback (TGF). Some of these vascular abnormalities may be mediated by increased cytosolic calcium content in afferent arterioles as a result of ischemia.

Tubuloglomerular Feedback

The role of TGF (see Chapter 2) in the setting of AKI may be partly beneficial because the resultant decrease in GFR limits sodium chloride delivery to damaged tubules. This in turn leads to reduced ATP-dependent tubular reabsorption of sodium, which protects against intracellular ATP depletion and thus kidney injury.

Tubular Epithelial Cell Injury and the Development of ATI

The tubular cell may be injured because of ischemia (depletion of cellular energy stores [ATP]) or from direct cytotoxic injury. Following acute kidney ischemia, tubular cell injury may also result from the restoration of RBF (reperfusion injury). Mediators of tubular cell injury include reactive oxygen species (ROS), intracellular calcium influx, nitric oxide, phospholipase A2, complement, and cell-mediated cytotoxicity.[3] Mitochondrial injury can be caused by ROS, depletion of antioxidants, and increased intracellular calcium. Disruption of mitochondrial function exacerbates cellular injury due to disrupted energy metabolism and release of proapoptotic proteins. Autophagy is a mechanism by which cells degrade self-proteins, and it is a central part of the cellular response to stress and injury. Experimental work has shown that autophagy is important for removal of damaged mitochondria and the recovery of tubular epithelial cells from ischemic injury. ROS may be derived from local sources (including xanthine oxidase and cyclooxygenases secondary to mitochondrial injury) or from infiltrating leukocytes. In models of ischemic ATN, a variety of methods that inhibit ROS protect against kidney injury. Hypoxia inducible factor (HIF) and downstream mediators such as heme oxygenase 1 may serve to protect cells against ischemic injury.[4]

Factors that affect the integrity and function of the renal tubular epithelial cells and contribute to the reduction in GFR include (Fig. 70.7):

1. *Cell death:* Most tubular cells undergo cell death by apoptosis rather than necrosis. In animal models, kidney injury is ameliorated using caspase inhibitors and p53 inhibitors that decrease apoptosis.[5]
2. *Disruption of actin cytoskeleton:* A characteristic feature of sublethally injured cells is the disruption of the actin cytoskeleton. Activation of the cysteine protease calpain (partly due to increased intracellular calcium) can degrade actin-binding proteins such as spectrin and ankyrin. This leads to abnormal translocation of Na^+K^+-ATPase and other proteins from the basolateral membrane to the cytoplasm or apical membranes. In the proximal tubular cell, this loss of polarity results in impaired proximal reabsorption of filtrate with resultant increased distal NaCl delivery, which activates tubuloglomerular feedback.
3. *Cast obstruction:* Tubular cells are attached to the tubular basement membrane by $\alpha 3\beta 1$ integrins, which recognize RGD (arginine-glycine-aspartate) sequences in matrix proteins. Disruption of the actin cytoskeleton results in movement of integrins from basolateral positions to the apical membrane, leading to impaired cell matrix adhesion and cell detachment. Many of these detached cells are still viable and can be cultured from urine of patients with ATN. Sloughed proximal tubular cells can bind to RGD sequences in Tamm-Horsfall protein, resulting in cast formation and intratubular obstruction. In models of ischemic AKI, the elevation in tubular pressures may be inhibited by synthetic RGD peptides mitigating the obstructive process. Both increased sodium concentration in the tubular lumen resulting from lack of sodium reabsorption or back-leak (see below), as well as acidic pH observed in ATI, favor dimerization of uromodulin and cast formation. Casts seen in ATI include granular casts, waxy casts, and renal tubular epithelial cell casts.

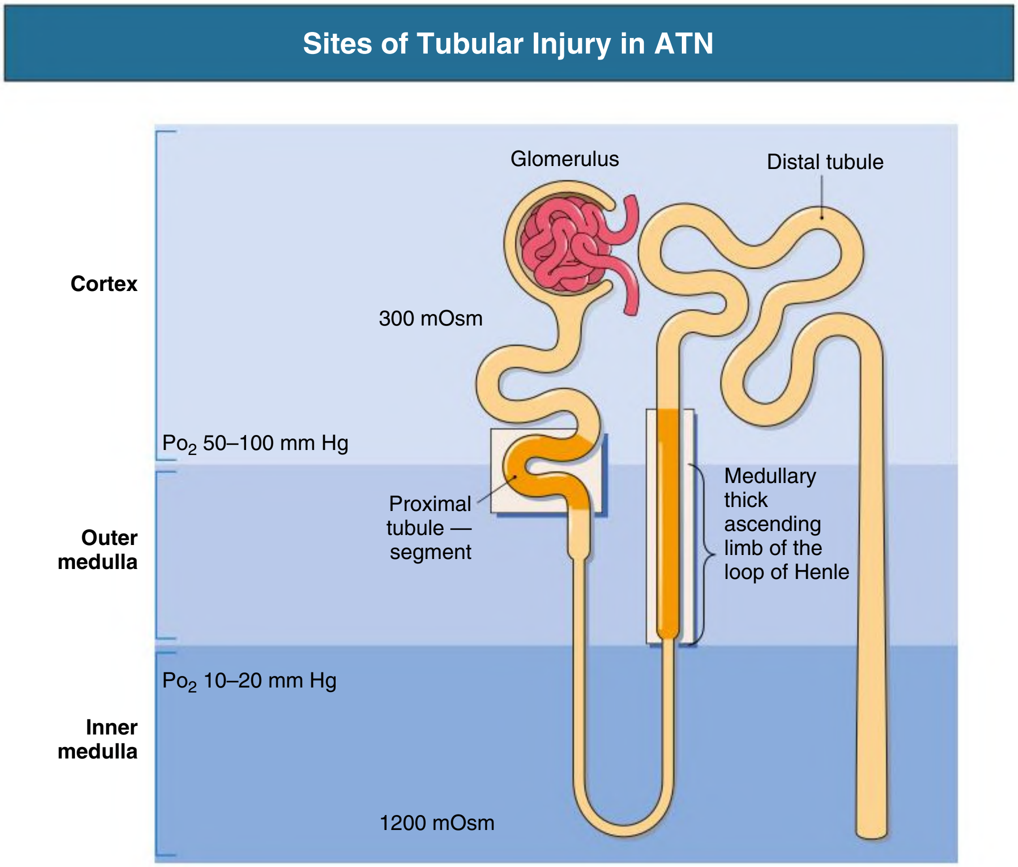

Fig. 70.5 Sites of Tubular Injury in Acute Tubular Necrosis (ATN). The S3 segment of the proximal tubule and the medullary thick ascending limb are particularly vulnerable to ischemic injury because of the combination of borderline oxygen supply and high metabolic demands.

4. *Back-leak:* The loss of adhesion molecules (E-cadherin) and tight junction proteins (ZO-1, occludin) weakens junctions between cells, allowing filtrate to leak back into the kidney interstitium. Although this does not alter the actual GFR at the level of the glomerulus, the net effect is a reduction in the measured GFR. Earlier dextran sieving experiments suggest only a modest effect of back-leak on the decrement of GFR in AKI (about 10%); however, in the kidney allograft with severe ATI, back-leak has been calculated to account for up to 50% of the GFR reduction.

Endothelial Cell Injury and the Development of ATN

Endothelial cell injury occurs partly as a result of acute kidney ischemia and oxidant injury.[3] Endothelial injury is characterized by cell swelling, upregulation of adhesion molecules (with recruitment of inflammatory neutrophils and monocytes), and impaired vasodilation (decreased endothelial nitric oxide synthase and vasodilatory prostaglandins) and may mediate some of the impaired autoregulation and intrarenal vasoconstriction described earlier. Endothelial injury within the peritubular capillaries (vasa rectae) may produce congestion in the outer medulla, exacerbating interstitial edema and worsening hypoxic injury to the S3 segment of the proximal tubule and the thick ascending loop of Henle.

Inflammatory Factors in The Development of ATN

Although ischemia causes direct kidney cytotoxicity, tissue inflammation during reperfusion also contributes to kidney injury and may cause some of the systemic effects of AKI. Components of both the innate and the adaptive immune systems contribute to the pathogenesis of ATN.[6] The innate immune system is activated by cellular injury and certain pattern recognition molecules. Toll-like receptor 2 (TLR2) and TLR4 are upregulated within the ischemic kidney, activated by molecules released from injured cells, and induce renal epithelial cells to produce chemokines. The complement system is also activated within the tubulointerstitium after ischemia/reperfusion, predominantly by the alternative pathway. It can directly induce nearby epithelial cells to produce proinflammatory cytokines (e.g., tumor necrosis factor-α [TNF-α], interleukin-6 [IL-6], IL-1β), and chemokines (e.g., MCP-1, IL-8, RANTES) that promote the infiltration of leukocytes and are also directly vasoactive.

A network of dendritic cells extends throughout the kidney interstitium, which help shape the inflammatory response within the kidney after ischemia/reperfusion, likely through their interactions with other inflammatory cell types. Neutrophils and mononuclear cells are seen in peritubular capillaries. Neutrophil activation and the release of proteases and ROS can exacerbate injury. By contrast, neutrophil depletion or the inhibition or genetic deletion of neutrophil adhesion molecules (ICAM-1) ameliorates injury in ischemic ATN. Monocytes infiltrate the kidney after reperfusion and differentiate into the M1 (proinflammatory) type exacerbating kidney injury after ischemia. These macrophages may later convert to an M2 (reparative) phenotype. Cells of the adaptive immune system, including T and B lymphocytes, also contribute to kidney injury in models of ATN, and their depletion ameliorates injury. It is not known whether these responses are antigen specific. Furthermore, some B and T cell subsets, such as T regulatory cells, help limit kidney injury.

AKI may have systemic effects on other organs.[7] The injured kidney may prime and activate leukocytes, which produce proinflammatory cytokines that can mediate remote organ injury (Fig. 70.8). The lungs may be particularly vulnerable from the combined effects of volume

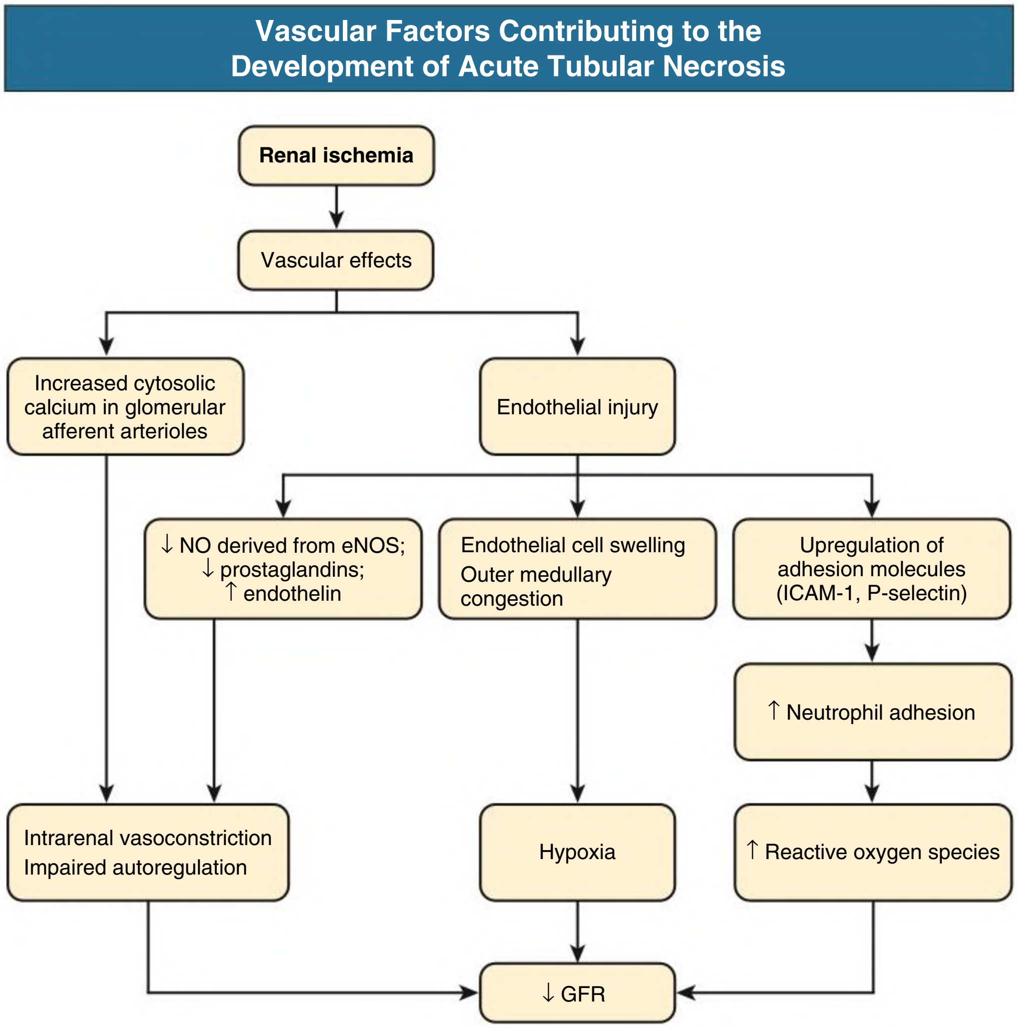

Fig. 70.6 **Vascular Factors Contributing to the Development of Acute Tubular Necrosis.** Renal vasoconstriction and endothelial injury promote kidney ischemia and tubular injury. *eNOS,* Endothelial nitric oxide synthase; *GFR,* glomerular filtration rate; *ICAM-1,* intercellular adhesion molecule 1; *NO,* nitric oxide.

overload, increased vascular permeability and proinflammatory environment. These effects may partly account for the increased mortality in patients with AKI (see Chapter 73).

Recovery Phase

Recovery from ATI requires the restoration of tubular cell number and coverage of denuded tubular basement membrane. Marked cell proliferation occurs in recovering human ATI (see Fig. 70.3). The restoration of tubular cell number is due to the dedifferentiation and proliferation of surviving tubular cells rather than from a mesenchymal stem cell source.[8] After tubular epithelial cell proliferation, the dedifferentiated cells must migrate to areas of denuded tubular basement membrane, attach to the basement membrane, and differentiate into mature polar tubular epithelial cells. The early inflammatory infiltrates of neutrophils and M1 monocytes are replaced by M2 monocytes, which support epithelial cell repair, after which their numbers decline by migration or apoptosis. When the injury process is persistent or severe, maladaptive repair may occur, leading to CKD.[9]

ACUTE CORTICAL NECROSIS

The same pathophysiological processes that lead to ATN can lead to bilateral acute cortical necrosis, but this only accounts for less than 2% of cases of ischemic ATN. If cortical necrosis is extensive, kidney failure often is irreversible. About 20% of the cases have reversible bilateral acute cortical necrosis.

In cortical necrosis, there is microvascular and glomerular thrombosis with extensive death of kidney tissue. The process is distinct from that seen with infarction caused by arterial occlusion because some blood supply to the medulla is preserved. The type of shock, in particular placental abruption, rather than severity seems to predict the risk for cortical necrosis. Other causes include thrombotic microangiopathy, snakebite envenomation, and trauma-related hemorrhagic shock.

The diagnosis of cortical necrosis is most often considered when recovery from AKI attributed to ATN is delayed. Contrast-enhanced computed tomography (CT) may reveal characteristic anatomic distortion of the kidneys (Fig. 70.9), but the use of radiocontrast in this setting has to be considered judiciously. Calcification detected radiologically is a late and unreliable sign. Contrast angiography may show patchy loss of perfusion. Kidney biopsy may not be informative because cortical necrosis may be patchy.

PATHOPHYSIOLOGY AND ETIOLOGY OF POSTRENAL AKI

Obstruction must be excluded in any patient with AKI because prompt intervention can result in improvement or complete recovery of kidney function (see Chapter 61). Most causes of obstructive nephropathy

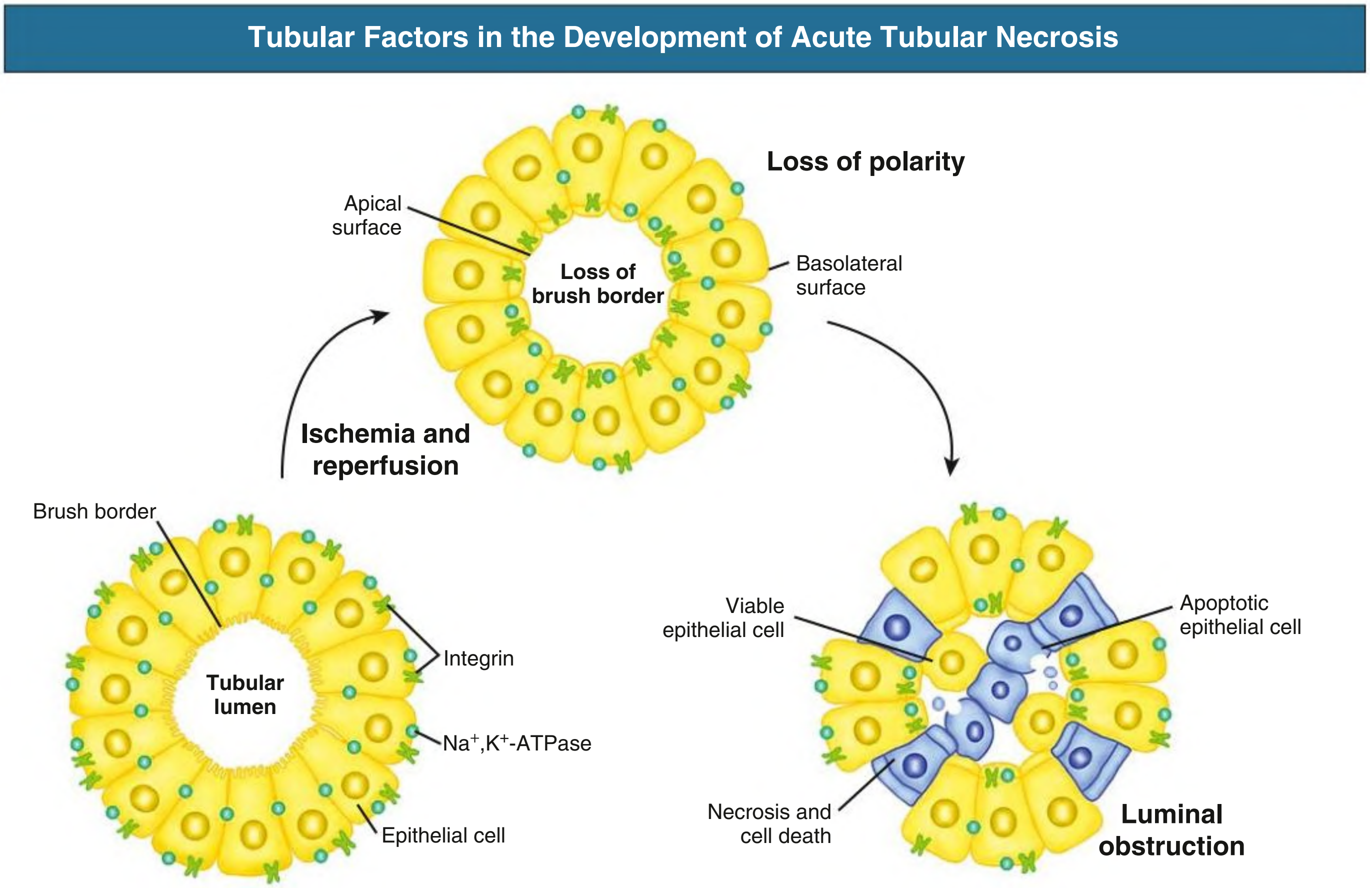

Fig. 70.7 Tubular Factors in the Development of Acute Tubular Necrosis. Loss of cell polarity results in weakening of cell-to-cell and cell matrix adhesion resulting in cast obstruction and back-leak of tubular fluid. (Modified from Schrier RW, Wang W, Poole B, Mitra A. Acute renal failure: definitions, diagnosis, pathogenesis, and therapy. *J Clin Invest.* 2004;114[1]:5–14.)

are amenable to therapy, and the prognosis is often good if treatment or intervention is early (Chapter 61).

NEPHROTOXIC AGENTS AND MECHANISMS OF TOXICITY

The identification and avoidance of nephrotoxic agents in AKI is critical because AKI may be rapidly reversible upon removal of the offending agent. The mechanisms of nephrotoxicity include alterations in kidney hemodynamics, induction of direct tubular injury, allergic reactions resulting in interstitial nephritis, and intratubular obstruction. The list is extensive, but the more common agents are presented in Fig. 70.10.

Nonsteroidal Antiinflammatory Drugs

NSAIDs commonly cause AKI in the community because of the large amounts of these drugs either prescribed or purchased over the counter. COX-2–specific NSAIDs have similar effects on kidney function as the nonselective NSAIDs and are not safer with respect to AKI. NSAID-related AKI is most often due to a hemodynamically mediated reduction in GFR that occurs in patients who are particularly dependent on vasodilatory prostaglandins to maintain kidney perfusion. These include elderly patients with atherosclerotic disease, volume-depleted patients, and those in sodium avid states such as cirrhosis, nephrotic syndrome, and congestive heart failure. This form of AKI is usually reversible in 2 to 7 days upon discontinuation of the drug and rarely occurs in otherwise healthy individuals. Concurrent use of diuretics, ACE inhibitors, ARBs, and calcineurin inhibitors has been associated with an increased incidence of NSAID-related AKI. Less frequently, NSAIDs induce ATN or, even more rarely, papillary necrosis. NSAIDs may also cause an acute interstitial nephritis (AIN), often with significant proteinuria (see Chapter 64). Other renal side effects of NSAIDs include fluid and electrolyte disturbances such as sodium retention exacerbating hypertension and congestive heart failure, hyponatremia, and hyperkalemia.

Acetaminophen (Paracetamol)

Isolated ATN with acetaminophen may occur in rare cases, but kidney injury is more typically associated with acute hepatitis. Kidney and liver toxicity usually occurs when more than 15 g have been taken, but in alcoholics, normal doses may be toxic. Acetaminophen is conjugated in the liver and undergoes renal excretion. Less than 5% undergoes metabolism by P-450 (CYP2E1) enzymes to form a toxic metabolite, N-acetylimidoquinone, which is inactivated by the thiol group of glutathione. With high levels of acetaminophen, glutathione becomes depleted, and N-acetylimidoquinone can bind to thiol groups on intracellular proteins, resulting in cell injury. Because glutathione is a major intracellular antioxidant, its loss may predispose to oxidative injury of the tubular cells.

Clinically, acute liver failure and ATN only begin once glutathione levels are depleted, and clinical manifestations usually present 3 to 4 days after ingestion. N-acetylcysteine, which substitutes for glutathione by providing a free thiol group, can be protective if administered early.

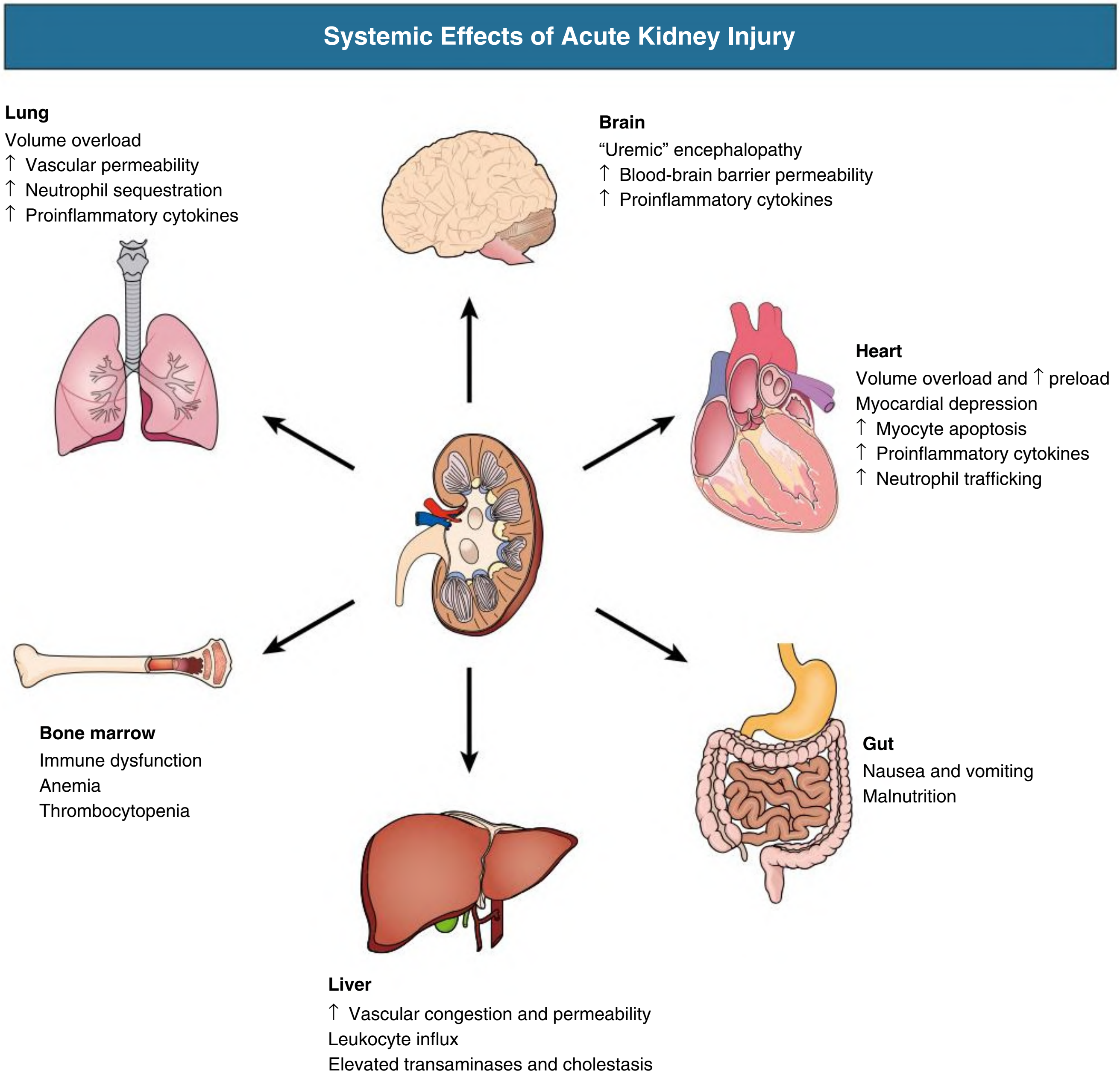

Fig. 70.8 Systemic Effects of Acute Kidney Injury. Acute kidney injury may contribute to remote organ injury, partly due to abnormalities in inflammatory and immune function from renal tubular dysfunction. (Modified from Grams ME, Rabb H. The distant organ effects of acute kidney injury. *Kidney Int.* 2012;81[10]:942–948.)

ACE Inhibitors and ARBs

Initiation of therapy with an ACE inhibitor or an ARB is commonly associated with a functional decrease in GFR that is not considered AKI. Indeed, it is well established that use of these medications is associated with improved kidney outcomes despite the initial drop in GFR that is often seen. ACE inhibitors and ARBs may also cause hemodynamically induced AKI in the setting of reduced kidney perfusion by impairing compensatory vasoconstriction of the efferent arteriole. These drugs may directly impair kidney perfusion by their antihypertensive effects. Patients in whom kidney perfusion is compromised due to dehydration, renovascular disease, or functionally impaired autoregulation are at particular risk for developing AKI after initiation of therapy. Patients chronically treated with ACE inhibitors or ARBs have an increased risk for postoperative kidney dysfunction, probably as a consequence of intraoperative hypotensive episodes.

Aminoglycosides

Aminoglycoside toxicity may occur if the dose is not adjusted to the GFR. Cationic amino groups (NH_3^+) on the drugs bind to anionic megalin on the brush border of proximal tubular epithelial cells. Following endocytosis aminoglycosides accumulate in proximal tubular cell lysosomes and can reach 100 to 1000 times their serum concentration. These drugs interfere with cellular energetics, impair intracellular phospholipases, and induce oxidative stress.[10]

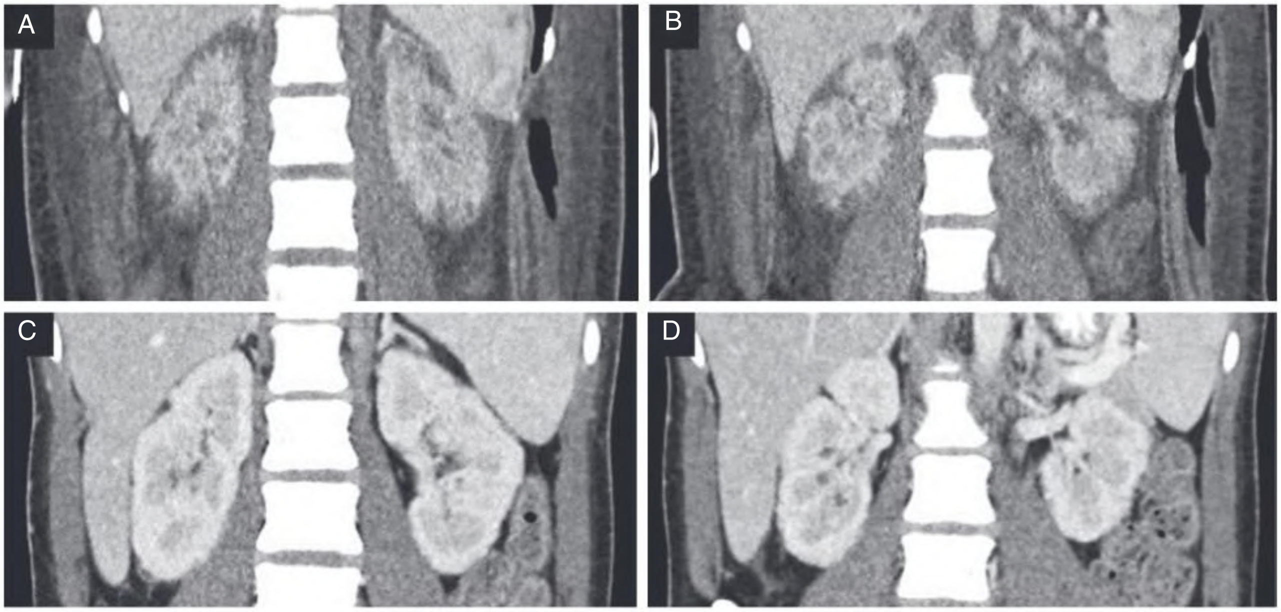

Fig. 70.9 Imaging Showing Bilateral Acute Cortical Necrosis of the Kidneys. (A–B) Contrast-enhanced computed tomography (CT) scanning coronal images disclose remarkable distortion of the anatomic appearance of both kidneys, including irregular margins, presence of hypodensity of the renal cortices and patchy medullary enhancement. (C–D) After 60 days, repeat CT scanning (coronal view) shows complete resolution of previously observed kidney abnormalities. (From Luo X, Cielo AG, and Velez JCQ. Abnormal imaging findings of the kidneys in a patient with shock. *Kidney360*. 2020;1[12]:1462–1463.)

Nonoliguric AKI usually occurs after 5 to 10 days of treatment with gentamicin. Involvement of distal tubular segments may produce polyuria, potassium, and magnesium wasting. The risk for AKI correlates with the accumulation of gentamicin in proximal tubular cells and is related to the daily dose and duration of therapy. Prolonged accumulation in proximal tubular cells may allow development of AKI even after the drug has been discontinued. Additional risk factors for gentamicin toxicity include increasing age, preexisting kidney disease, hypotension, concurrent liver disease, sepsis, coadministration of vancomycin, and concurrent nephrotoxins. Aminoglycoside serum levels should be monitored to minimize nephrotoxicity. When possible, the drug should be administered in a single daily total dose, which leads to lower renal proximal tubular cell accumulation. Gentamicin, tobramycin, and netilmicin appear to have similar nephrotoxic effects. Amikacin, which has fewer amino groups per molecule, and plazomicin, a next-generation aminoglycoside, may be less nephrotoxic.

Vancomycin

The extent to which vancomycin is directly nephrotoxic is controversial.[11] The recommendation for higher vancomycin trough levels to target methicillin-resistant *Staphylococcus aureus* has led to recognition that vancomycin can be nephrotoxic. Vancomycin is excreted primarily by glomerular filtration, but accumulation in proximal tubule cells via basolateral secretion is thought to underlie nephrotoxicity. Experimentally, high-dose vancomycin causes oxidative stress and triggers intrarenal apoptotic pathways. In humans, high initial vancomycin trough levels greater than 15 μg/mL have been associated with nephrotoxicity in a graded fashion. Additional risk factors include total dose of more than 4 g daily, long duration of therapy, concurrent nephrotoxin or diuretic exposure, and critical illness. Vancomycin coadministration with piperacillin-tazobactam appears to increase risk for AKI, though the mechanism for this is unclear.[12] A precipitous rise in serum creatinine could be observed in a subset of cases of vancomycin-associated AKI.[13] ATN is the predominant lesion seen in experimental models of vancomycin nephrotoxicity, but case reports of human biopsies have shown both interstitial nephritis and ATN. Vancomycin can cause a drug reaction with eosinophilia and systemic symptoms (DRESS syndrome) with inflammatory kidney injury. Treatment is generally conservative. High-flux hemodialysis can be used for drug removal when levels are very high. Kidney injury from vancomycin is generally reversible.

Amphotericin B

This polyene macrolide antibiotic binds to sterols in the cell membranes of both fungal walls (ergosterol) and mammalian (cholesterol) cell membranes, resulting in the formation of aqueous pores that increase membrane permeability. Within the renal tubular cell, the subsequent sodium influx leads to increased Na^+,K^+-ATPase activity and depletion of cellular energy stores.[14] Additionally, the standard amphotericin B formulation is suspended in the bile salt deoxycholate, which has a detergent effect on cell membranes. Nephrotoxicity relates to cumulative dosage, usually occurring after administration of 2 to 3 grams.

Early signs of nephrotoxicity include a loss of urine-concentrating ability, followed by a decrease in GFR. Hypokalemia and hypomagnesemia due to distal tubular wasting are common. A distal renal tubular acidosis may be present due to proton back-leak in the cortical collecting duct.

Prevention of nephrotoxicity requires the maintenance of high urine flow rates by saline loading during amphotericin administration. The more expensive liposomal amphotericin B preparations reduce the incidence of AKI by approximately 50%. As amphotericin B binds more avidly to fungal ergosterol than to cholesterol, delivering the drug as a cholesterol liposome diminishes binding to tubular epithelial cell membranes without altering fungicidal activity. Additionally,

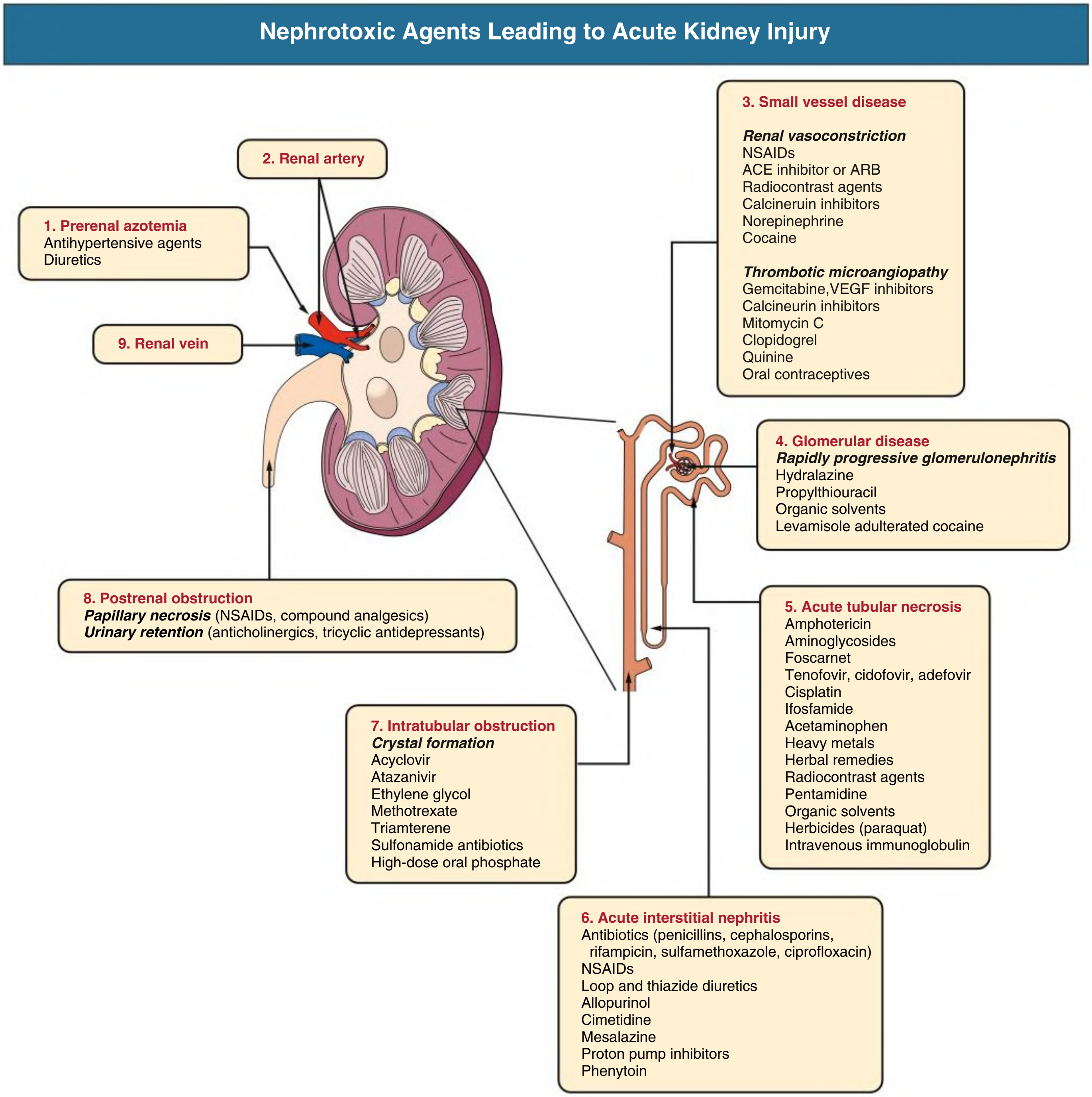

Fig. 70.10 Common nephrotoxic agents leading to acute tubular necrosis. *ACE,* Angiotensin-converting enzyme; *ARB,* angiotensin receptor blocker; *NSAID,* nonsteroidal antiinflammatory drug; *VEGF,* vascular endothelial growth factor.

liposomal preparations do not contain deoxycholate. Amphotericin B–induced AKI is usually reversible with discontinuation of the drug, although distal tubular injury as manifested by magnesium wasting may persist.

Antiviral Therapy

Acyclovir

Nephrotoxicity is typically seen after intravenous (IV) acyclovir administration and may be due to direct tubular cell toxicity and the formation of intratubular, birefringent needle-shaped acyclovir crystals. However, kidney biopsy data suggest that AIN may be the predominant mechanism of toxicity.

Oliguric AKI typically occurs within a few days of treatment and may be associated with abdominal or loin pain. High serum levels of acyclovir due to decreased kidney clearance may produce neurologic toxicity. The AKI is usually mild and recovers on stopping the drug. Maintaining a high urine flow rate and avoiding IV bolus administration of acyclovir may be preventative.

Tenofovir

Tenofovir is a nucleoside reverse transcriptase inhibitor used to treat both HIV and hepatitis B infection. The prodrug, tenofovir disoproxil fumarate (TDF), is the most widely used therapy for HIV in the world. Tenofovir is secreted into proximal tubule cells via organic anion transporters (OATs), where it can interfere with mitochondrial DNA synthesis and upset the energy supply, resulting in characteristic enlarged, dysmorphic mitochondria. Manifestations of tenofovir nephrotoxicity include subclinical tubular defects (e.g., normoglycemic glycosuria), Fanconi syndrome, ATN, and CKD.[15] Kidney manifestations can develop within weeks of drug initiation or can occur after years, but reversibility is common. Risk factors include preexisting CKD, advanced age, low CD4 count, and total dose and duration of TDF use. HIV patients exhibit a variety of kidney diseases (see Chapter 58), and biopsy should be strongly considered when kidney dysfunction does not improve after drug cessation or when resistance patterns make tenofovir critical for patient care. Tenofovir alafenamide (TAF) is a newer prodrug of tenofovir, the use of which has increased in recent years such that it is now included in many first-line drug regimens. TAF concentrates in mononuclear cells, resulting in lower plasma concentrations and reduced drug delivery to the kidney. TAF is less nephrotoxic than TDF, but cases of TAF-associated AKI have been reported.[16]

Other Antiviral Agents

Among antivirals used to treat HIV infection, both cobicistat and dolutegravir inhibit proximal tubular secretion of creatinine, causing a false elevation in serum creatinine. A small rise in creatinine is considered acceptable and generally occurs within 2 weeks of starting these drugs. Atazanavir, a protease inhibitor, can cause urolithiasis, crystalline-induced kidney injury, AIN, or a granulomatous interstitial nephritis. Acute presentations are largely reversible, but long-term use can lead to CKD.[17]

Adefovir, a second-line drug used to treat hepatitis B, was frequently nephrotoxic when prescribed at high doses, but current low-dose regimens appear safe for most patients. Foscarnet and cidofovir are well-recognized nephrotoxic antivirals.

Sodium-Glucose Cotransporter-2 Inhibitors

Sodium-glucose cotransporter-2 (SGLT2) inhibitors can be associated with a functional decrease in GFR that is likely the result of direct renovascular effects, similar to that seen with initiation of ACE inhibitors and ARBs, and is not considered to represent AKI. In fact, the risk of AKI as a direct result of treatment with SGLT2 inhibitors is low.[18] There are rare case reports of AKI due to osmotic nephrosis following initiation of an SGLT2 inhibitor. The use of SGLT2 inhibitors in CKD is covered in Chapters 32 and 33.

Immunosuppressive Agents

Calcineurin Inhibitors

Cyclosporine and tacrolimus may cause AKI due to afferent arteriolar vasoconstriction, partly mediated by endothelin. This is usually reversible upon dose reduction. Persistent injury may lead to chronic interstitial fibrosis in a striped pattern along medullary rays reflecting both the ischemic nature of the insult, as well as the development of arteriolar hyalinosis. Associated clinical features include hypertension, hyperkalemia, hyperuricemia, and wasting of phosphorus and magnesium from tubular injury. Calcineurin inhibitors also cause reversible tubular dysfunction, and are associated with the development of thrombotic microangiopathy, likely due to their effects on the endothelium (see Chapter 30).

Other Immunosuppressive Agents

The monoclonal anti-CD3 antibody (OKT3) or polyclonal antilymphocyte and antithymocyte preparations (ALG, ATG) may cause a first-dose cytokine release syndrome and prerenal azotemia secondary to capillary leak. Intravenous immunoglobulin (IVIG) can cause AKI, which may be partly mediated by the high sucrose concentration in these products. Tubular uptake of sucrose may result in osmotic cell swelling and injury. Although it does not typically cause AKI, sirolimus delays the recovery from AKI in kidney transplant patients with delayed graft function.

Anticoagulation-Related Nephropathy

AKI has been described in patients taking anticoagulants who develop an acute rise in the international normalized ratio (INR), usually to greater than 3. Most cases of anticoagulation-related nephropathy (ARN) appear within the first 8 weeks of initiating therapy in patients with underlying CKD. AKI results from glomerular hematuria with obstruction of renal tubules by red blood cell casts. The risk of ARN is thought to be highest in patients treated with warfarin, but cases in patients treated with direct acting oral anticoagulants have also been reported.[19]

Acute Phosphate Nephropathy

Oral sodium phosphate has been widely used as a bowel preparation for colonoscopy procedures, but recent cases of AKI have limited the use of this purgative.[20] AKI associated with oral sodium phosphate is believed to be caused by phosphaturia and acute calcium-phosphate deposition within the renal tubules. Risk factors for this condition include older age, volume depletion, and underlying CKD. A similar phenomenon has been reported with use of sodium phosphate–containing enemas.

Drugs of Abuse

AKI is a common condition in those who abuse drugs and may be due to nephrotoxicity of the drug, coexistent viral infection (HIV, HCV), sepsis, infective endocarditis, or rhabdomyolysis.[21]

Cocaine induces intense vasoconstriction, which may lead to AKI due to severe hypertension or rhabdomyolysis. Mechanisms for rhabdomyolysis include coma and pressure necrosis, vasospasm leading to ischemic muscle injury, and adrenergic stimulation with hyperpyrexia leading to increased cellular metabolism. Cocaine may also exert direct toxic effects on the myocyte.[22] Most cocaine entering the United States is contaminated by levamisole, an immunomodulator and antihelminthic agent. In humans, levamisole has been reported to cause an antineutrophil cytoplasmic antibody (ANCA)-positive vasculitis with a necrotizing crescentic glomerulonephritis and prominent skin lesions.

Other illicit drugs associated with AKI include opiates (coma-associated, pressure-induced rhabdomyolysis); phencyclidine (rhabdomyolysis secondary to hyperpyrexia and vasoconstriction); and methamphetamines (AKI secondary to rhabdomyolysis, AIN, or acute necrotizing angiitis). Recently, synthetic cannabinoids have been implicated as a cause of AKI, although the culprit component is unclear. When available, kidney biopsy has shown ATN and more rarely AIN. In most cases, no other cause for AKI could be identified.[21]

Ethylene Glycol

Ethylene glycol, found in antifreeze, is a cause of both deliberate and accidental overdose. It is rapidly metabolized by alcohol dehydrogenase to glycoaldehyde and glyoxylate, which are toxic to tubular cells. Further metabolism generates oxalic acid, which can precipitate, leading to intratubular obstruction.

The diagnosis is suggested by the presence of a severe anion gap metabolic acidosis and an elevated serum osmolal gap. Calcium oxalate crystals are often seen on the urine microscopy (see Chapter 4, Fig. 4.4). Management includes inhibition of alcohol dehydrogenase with fomepizole, or IV ethanol if this agent is not available. Although there is no specific ethylene glycol level above which extracorporeal removal is mandated, hemodialysis is the quickest way to remove both parent alcohol and toxic metabolites. Methanol intoxication may present with similar metabolic abnormalities but rarely causes AKI (see Chapter 13).

Occupational Toxins

Heavy Metals

Lead intoxication usually causes a chronic nephropathy (see Chapter 65). Rarely, ATI occurs that may be associated with Fanconi syndrome. AKI may also occur in cadmium and mercury poisoning.

Organic Solvents

Organic solvents may cause ATI due to peroxidation of membrane lipids. Subacute kidney failure due to anti–glomerular basement membrane (anti-GBM) antibody disease has also been reported with exposure to halogenated hydrocarbons.

Herbal Remedies

AKI has been reported with the use of herbal remedies; however, nephrotoxicity is most often due to impurities found in unregulated formulations of these substances (e.g., lead, NSAIDs) and not the herbs specifically. Some herbs used in traditional African medicine (e.g., Cape aloes, *Callilepis laureola*) are common causes of AKI in parts of Africa. Aristolochic acid (found in certain traditional Chinese medicines) can cause subacute kidney failure (see Chapters 65 and 79).

Contrast-Induced Nephropathy

It is generally agreed that iodinated contrast can be nephrotoxic but that the risk of AKI with contrast administration is frequently overstated. AKI typically occurs in patients with underlying CKD and is rarely seen in patients with preserved estimated glomerular filtration rate (eGFR). In hospitalized patients, it is often difficult to determine whether a contrast CT is the primary cause of AKI, as imaging is often obtained in the setting of other potential kidney insults (e.g., infection, antibiotics). Recent studies have questioned the link of IV contrast CT with nephrotoxicity, suggesting minimal risk of AKI when low or isoosmolar contrast is administered intravenously to patients with eGFR greater than 30 mL/min/1.73 m^2.[23-25] Indeed, in these studies, the risks of AKI associated with contrast administration were similar among well-matched hospitalized patients who received noncontrast CT scans.

Intraarterial contrast administration, required for cardiac catheterization or renal angiography, poses a greater risk for AKI because a larger dose of iodinated material is required and is then delivered to the renal arteries in a concentrated manner. Risk factors for the development of AKI from contrast nephropathy include diabetic nephropathy, advanced age (>75 years), congestive heart failure, volume depletion, hyperuricemia, and high or repetitive doses of radiocontrast agents. Concurrent use of NSAIDs, ACE inhibitors, or diuretics may increase the risk (see Chapter 74).

Both kidney ischemia and direct tubular epithelial cell toxicity are implicated in the pathogenesis of contrast nephrotoxicity. Typically, a biphasic hemodynamic response is seen. Initial vasodilation (lasting a few seconds to minutes) is followed by more prolonged renal vasoconstriction. The resultant medullary hypoxia may be exacerbated by low flow in the vasa recta as a result of a contrast-induced rise in blood viscosity. An osmotic diuresis, leading to increased sodium delivery to the medullary thick ascending loop, may result in increased oxygen consumption for sodium reabsorption. Uricosuria, as well as a hyperosmolar activation of the aldose reductase-fructokinase system in the kidney, has also been proposed to play a role. Radiocontrast agents also cause direct tubular epithelial cell injury. Human studies have demonstrated low-molecular-weight proteinuria, suggestive of proximal tubular injury, partly mediated by ROS. The administration of antioxidants ameliorates contrast nephrotoxicity in animals.

OTHER SPECIFIC ETIOLOGIES OF AKI

Heme Pigment Nephropathy

Heme pigment nephropathy is a common cause of AKI and is usually secondary to the breakdown of muscle fibers (rhabdomyolysis), which release potentially nephrotoxic intracellular contents (particularly myoglobin) into the systemic circulation. Less commonly, heme pigment nephropathy may occur due to massive intravascular hemolysis. Prevention and therapy of heme pigment nephropathy are discussed in Chapter 74.

Causes of Rhabdomyolysis

Muscle trauma is the most common cause of rhabdomyolysis. The initial description was by Bywaters and Beall during the bombing of London in World War II.[26] Other common causes of muscle injury include marked exercise, seizures, pressure necrosis secondary to coma, substance abuse, and limb ischemia (Table 70.1). In skeletal muscles confined to rigid compartments, cell swelling after injury may result in increased intracompartmental pressures that can impair local microvascular circulation and lead to compartment syndrome (Fig. 70.11). In the patient with substance or alcohol abuse, rhabdomyolysis is often multifactorial. Contributing causes include pressure necrosis from coma (“found down”), direct myotoxicity from ethanol, seizures, and electrolyte abnormalities (hypokalemia and hypophosphatemia). Therapy with statins may be associated with rhabdomyolysis, especially when fibrates, cyclosporine, or erythromycin are used concurrently. The risk of rhabdomyolysis appears to be highest with simvastatin and lower with newer statins such as atorvastatin and rosuvastatin. Familial myopathies such as McArdle syndrome and carnitine palmitoyl transferase deficiency should be suspected in patients with

TABLE 70.1 Causes of Rhabdomyolysis

Muscle injury/ischemia	Trauma; pressure necrosis, electric shock, burns, acute vascular disease
Myofiber exhaustion	Seizures, excessive exercise, heat exhaustion
Toxins	Alcohol, cocaine, heroin, amphetamines, ecstasy, phencyclidine, snakebite
Drugs	Statins, fibrates, zidovudine, neuroleptic malignant syndrome, azathioprine, theophylline, lithium, diuretics
Electrolyte disorders	Hypophosphatemia, hypokalemia, hyperosmolar states
Infections	Viral: Influenza, HIV, coxsackievirus, Epstein-Barr virus, COVID-19 Bacterial: *Legionella, Francisella, Streptococcus pneumoniae, Salmonella, Staphylococcus aureus*
Familial	McArdle disease, carnitine palmitoyl transferase deficiency, malignant hyperthermia
Other	Hypothyroidism, polymyositis, dermatomyositis

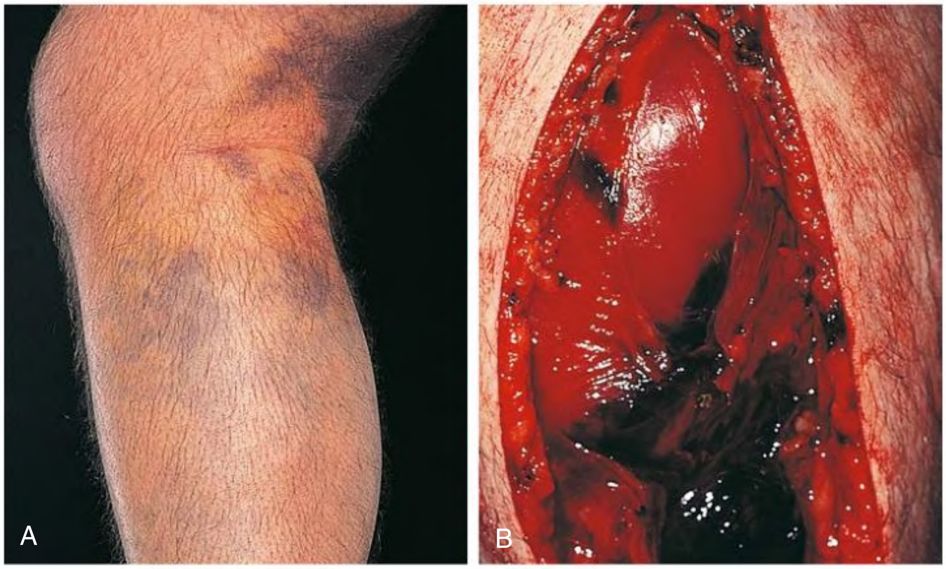

Fig. 70.11 **Compartment Syndrome.** (A) Severe calf swelling due to anterior and posterior compartment syndromes after ischemia-reperfusion. (B) Appearance after emergency fasciotomy: note edematous muscle and hematoma. (Courtesy Mr. M.J. Allen, FRCS.)

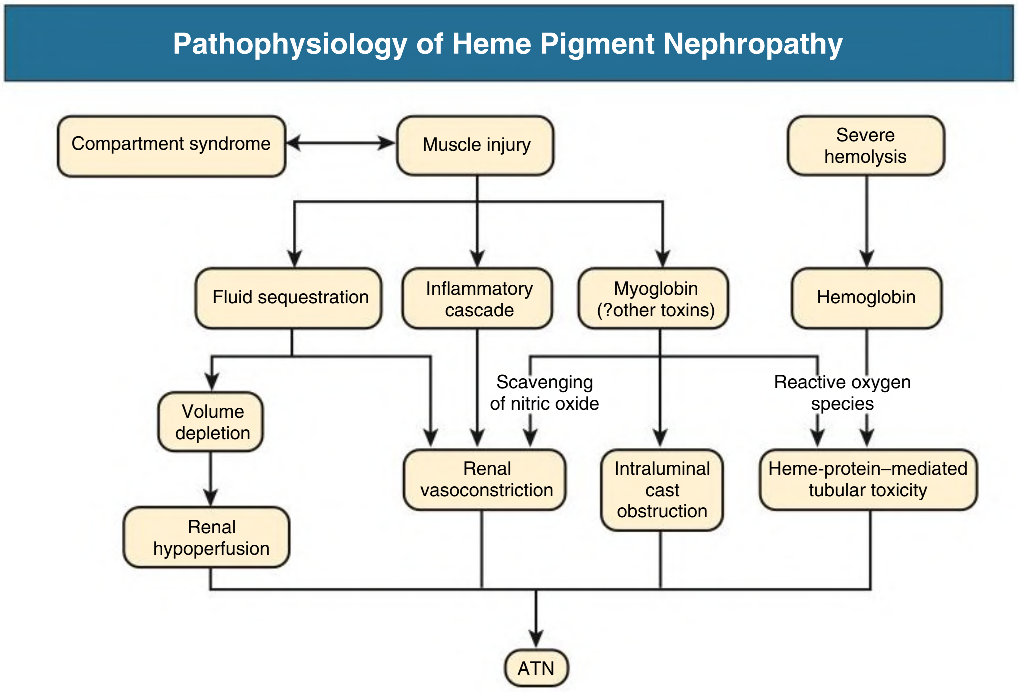

Fig. 70.12 **Pathophysiology of Heme Pigment Nephropathy.** *ATN,* Acute tubular necrosis.

recurrent episodes of rhabdomyolysis associated with muscle pain, positive family history, onset in childhood, and the absence of other identifiable causes. In developing countries, ingestion of hair dye containing paraphenylene-diamine as a means of self-harm may cause AKI secondary to rhabdomyolysis. Snake and spider bites, bee stings and venomous caterpillar bites may all cause rhabdomyolysis (see Chapter 71). There have been several reports of rhabdomyolysis associated with COVID-19 infection and recreational exposure to bath salts (3,4-methylenedioxypyrovalerone).

Causes of Hemoglobinuria

Intravascular hemolysis results in circulating free hemoglobin. If the hemolysis is mild, the released hemoglobin is bound by circulating haptoglobin. With massive hemolysis, however, haptoglobin stores become exhausted. Hemoglobin (69 kDa) then dissociates into α-β dimers (34 kDa), which are small enough to be filtered, resulting in hemoglobinuria, hemoglobin cast formation, heme uptake by proximal tubular cells, ATN, and filtration failure. Causes of hemoglobinuric AKI include incompatible blood transfusion, autoimmune hemolytic anemia, malaria (blackwater fever), glucose-6-phosphate dehydrogenase deficiency, paroxysmal nocturnal hemoglobinuria, march hemoglobinuria, and toxins (dapsone, venoms).

Pathogenesis of Heme Pigment Nephropathy

The kidney injury is due to a combination of factors including volume depletion, renal vasoconstriction, direct heme-protein–mediated cytotoxicity, and intraluminal cast formation (Fig. 70.12).[27] Volume depletion is often prominent in patients with rhabdomyolysis owing to the sequestration of up to 15 to 20 L of fluid in injured muscle. Volume depletion activates the sympathetic nervous system and renin-angiotensin system, resulting in renal vasoconstriction. This may be exacerbated by the scavenging of nitric oxide by circulating heme proteins.

Myoglobin (17 kDa) is freely filtered at the glomerulus and is toxic to tubular epithelial cells. The heme center of myoglobin may directly induce lipid peroxidation and kidney injury, and liberated free iron catalyzes the formation of hydroxyl radical through the Fenton reaction, inducing free radical–mediated injury. It has recently been demonstrated that myoglobin released from damaged skeletal muscle activates platelets, which, in turn, induce macrophages in the kidney to produce macrophage extracellular traps (METs).[28] METs have been implicated in the development of kidney injury in mice, though it remains unclear exactly which components of METs are responsible. Renoprotection has been demonstrated in animal models with free iron scavengers and various antioxidants. Finally, the precipitation of myoglobin with uromodulin (Tamm-Horsfall protein) and sloughed proximal tubular cells may result in obstructing casts in the distal nephron, especially when tubular flow rates are low due to volume depletion. The binding of myoglobin to uromodulin is enhanced in acidic urine.

Atheroembolic Kidney Disease

This underrecognized condition occurs predominantly in older patients with atherosclerotic vascular disease (see also Chapter 43). It can occur spontaneously but is most commonly precipitated by arteriography, vascular surgery, thrombolysis (streptokinase and tissue plasminogen activator), or anticoagulation. Destabilization of atherosclerotic plaques primarily in the aorta above the level of the renal arteries results in showers of cholesterol that lodge in small arteries in the kidneys (see Fig. 43.12) and the lower extremities (see Fig. 43.11). Characteristic needle-shaped clefts may be seen on kidney or skin biopsy, denoting the localization of cholesterol plaques before dissolution during tissue fixation (Fig. 43.12). The cholesterol emboli produce an inflammatory reaction and occlusion of the vasculature.

Renal Artery or Vein Occlusion

AKI can be caused by bilateral renal artery occlusion, or unilateral occlusion in the setting of a single functioning kidney (see also Chapter 43). Thrombosis or embolization (noncholesterol) of the renal artery or its intrarenal branches are more common in elderly patients. Atrial fibrillation is an important risk factor for renal emboli. Aortic dissection may also progress to occlude the renal arteries. Renal vein thrombosis most commonly occurs in the setting of nephrotic syndrome and rarely may cause AKI if bilateral.

Acute Interstitial Nephritis

This is most commonly a drug-induced phenomenon and is an important differential diagnosis in AKI because removal of the offending agent can result in reversal of the condition. Less commonly, interstitial nephritis may be due to infection or immune-mediated diseases (see Chapter 64).

Thrombotic Microangiopathy

Thrombotic microangiopathy (TMA) should be considered when a patient presents with AKI and thrombocytopenia, although the condition may occur in the absence of a low platelet count (see also Chapter 30).

TABLE 70.2 Causes of Pulmonary-Renal Syndrome

Anti-GBM Disease (Goodpasture)	
Systemic vasculitis	ANCA associated: • Granulomatosis with polyangiitis • Microscopic polyangiitis • Eosinophilic granulomatosis with polyangiitis • Drugs (penicillamine, hydralazine, propylthiouracil) Immune complex disease: • Lupus • IgA vasculitis • Mixed cryoglobulinemia • Rheumatoid vasculitis
Infection	Severe bacterial pneumonia, postinfectious glomerulonephritis, legionella, leptospirosis, hantavirus, infective endocarditis
Pulmonary edema and AKI	Volume overload in severe left ventricular failure
Multiorgan failure	Acute respiratory distress syndrome and AKI
Other	Paraquat poisoning, renal vein or inferior vena cava thrombosis with pulmonary emboli

AKI, Acute kidney injury; *anti-GBM,* anti–glomerular basement membrane; *ANCA,* antineutrophil cytoplasmic antibody; *IgA,* immunoglobulin A.

Glomerular Disease

All types of glomerular disease (see Section 4) may present with AKI, but it is more commonly seen with forms of acute glomerulonephritis such as postinfectious glomerulonephritis, ANCA-associated small-vessel vasculitis, anti-GBM disease, lupus nephritis, and immunoglobulin A nephropathy. In nephrotic syndrome, AKI may occur due to volume depletion from diuretic use, superimposed ATN, renal vein thrombosis, or rarely from superimposed acute glomerulonephritis (e.g., anti-GBM disease in membranous nephropathy). Minimal change disease in adults is the most common nephrotic disorder associated with AKI.

Pulmonary-Renal Syndromes

The term pulmonary-renal syndrome usually describes the presence of pulmonary hemorrhage in a patient with acute glomerulonephritis. It can be caused by anti-GBM disease (Goodpasture syndrome), systemic vasculitis, or systemic lupus erythematosus (see Chapters 25–27). Patients with pulmonary-renal syndrome require urgent evaluation, and specific testing for these diseases should be pursued. A similar clinical presentation may occur in patients with pulmonary infection and AKI, or in the setting of pulmonary edema secondary to volume overload from AKI. Other conditions that may masquerade as a pulmonary-renal syndrome are outlined in Table 70.2.

SPECIFIC CLINICAL SITUATIONS

AKI in the Patient With Sepsis

Sepsis accounts for up to 50% of cases of AKI in intensive care unit patients, and septic AKI requiring renal replacement therapy is associated with mortality rates as high as 50% to 60%.[29] The traditional model of hypotension and vasoconstriction leading to ischemic ATN has been challenged because in early sepsis, renal blood flow is often normal or even increased. Moreover, the severe drop in GFR observed with sepsis is out of proportion to the relatively mild histologic injury seen on kidney biopsies, and on autopsy only 22% of patients with sepsis-associated AKI had histologic features of ATN.[30]

The pathogenesis of sepsis-associated AKI is mediated by molecules released from pathogens (lipopolysaccharide, flagellin, lipoteichoic acid, DNA), or injured cells, which activate the innate immune system.[31,32] This leads to the activation of a wide range of cellular and humoral mediator systems, including the cytokine cascade (TNF-a, IL-1β, IL-6); the complement, coagulation, and fibrinolytic systems; increased oxidative stress; and the release of mediators such as eicosanoids, platelet-activating factor, endothelin-1, and nitric oxide. Renal endothelial cells can be damaged in the procoagulant milieu and upregulate their expression of adhesion molecules, further amplifying the immune response. Disruption of the renal microcirculation ensues, with local areas of cortical ischemia despite maintenance of normal renal arterial blood flow. Ligation of toll-like receptors on tubular cells (TLR-4) leads to mitochondrial dysfunction, oxidative stress, and severe apoptosis. The mechanism for the abrupt GFR decline in early sepsis likely involves *efferent* arteriolar vasodilation. Therapies that promote efferent arteriolar vasoconstriction may be beneficial. Treatment with arginine vasopressin, which constricts the efferent more than the afferent arteriole, was associated with reduced progression to the most severe category of AKI in patients with septic shock.[33] Administration of large volumes of IV fluid is common in the early resuscitation phase of sepsis. Isotonic crystalloid solutions with supraphysiologic concentrations of chloride (e.g., normal saline) are associated with an increased incidence of AKI compared with balanced crystalloid solutions (e.g., lactated ringers and plasmalyte), likely due to intense afferent arteriolar vasoconstriction resulting from increased chloride delivery to the macula densa. Sepsis may lead to multiorgan failure, and affected patients may experience repeated episodes of AKI due to nephrotoxic drug exposure or hospital-acquired infections. Maladaptive repair processes can lead to CKD following AKI in sepsis.

AKI in COVID-19

AKI is a common complication of COVID-19; ATN is the predominant mechanism of injury. Cases of collapsing focal segmental glomerulosclerosis have also been reported; these patients typically present with AKI in addition to nephrotic range proteinuria. Kidney disease in COVID-19 is discussed further in Chapter 59.

AKI in the Trauma Patient

AKI significantly increases mortality among those with severe trauma.[34] Mechanisms for kidney injury include rhabdomyolysis (earthquake victims, crush injuries, burns), ATN from hypovolemic shock, or abdominal compartment syndrome due to massive hemorrhage or aggressive hydration. Direct kidney injury may result from penetrating (gunshot, stab wound) or more commonly blunt trauma (fall, motor vehicle collision, assault, sports injury). Proper diagnosis of these injuries requires a contrast-enhanced CT with delayed imaging, otherwise late extravasation from the kidney pelvis or ureters may be missed. Management is generally conservative, but emergent nephrectomy may be required.

AKI in the Postoperative Patient

Postoperative AKI is commonly due to perioperative hemodynamic instability, volume depletion, and/or nephrotoxin exposure.

After Cardiac Surgery

Risk factors for postoperative AKI include duration of cardiac bypass, preoperative kidney function, hyperuricemia, age, diabetes, valvular surgery, blood transfusions, and poor cardiac function.[35] The surgery is often performed with the patient cooled to less than 30°C to protect cells against ischemic injury; however, systemic hypothermia may cause intravascular coagulation. Aortic instrumentation and cross-clamping may lead to renal atheroembolism. Cardiac bypass causes loss of pulsatile flow and exposure of blood to a nonendothelialized surface, resulting in activation of neutrophils, platelets, complement, and fibrinolytic systems. Significant hemolysis and hemoglobinuria may also occur. Perioperative myocardial infarction or left ventricular dysfunction may impair kidney perfusion postoperatively, although the low cardiac output is often transient (myocardial stunning) and recovers within 24 to 48 hours. Atrial fibrillation is common and may be associated with peripheral embolization. Off-pump coronary artery bypass operations may have a lower risk of AKI than surgeries involving cardiopulmonary bypass.

After Vascular Surgery

AKI is common after abdominal aortic aneurysm repair. The risk appears to be lower with endovascular versus open repair.[36] Mechanisms for AKI include ischemic ATN due to prolonged aortic cross-clamping, renal artery thromboembolism or dissection, and cholesterol atheroemboli. Additional complications specific to endovascular repair include contrast nephropathy and kidney ischemia due to endograft malpositioning or migration. In patients with peripheral vascular disease, there is often ischemic kidney disease. Preoperative reduction in eGFR is the strongest predictor of the risk for postoperative AKI.

Abdominal Compartment Syndrome

Markedly raised intraabdominal pressures (>20 mm Hg) may occur after trauma, after abdominal surgery, or secondary to massive fluid resuscitation and can cause AKI. Particularly vulnerable are patients with low mean arterial pressures, as this further compromises abdominal perfusion pressure.[37] The multifactorial mechanisms of AKI likely include increased kidney venous pressure and vascular resistance, as well as a drop in cardiac output from decreased venous return and reduced ventricular compliance. Intraabdominal pressures are best estimated by measurement of intravesical (bladder) pressures. Efforts to reduce intraabdominal pressures, including paracentesis, nasogastric suction, ultrafiltration, or surgical decompression, may improve kidney function.

AKI and Liver Disease

AKI is common in patients with cirrhosis. The differential diagnosis is typically between prerenal AKI, ATN, and hepatorenal syndrome. The pathophysiology of hepatorenal syndrome is discussed in Chapter 76. Assessment of intravascular volume status can be difficult, and a therapeutic trial of volume replacement is typically undertaken. Risk factors for AKI in this population include hypovolemia, GI bleeding, and infection (particularly spontaneous bacterial peritonitis). Severe malnutrition and decreased muscle mass may be masked by the presence of edema; in these settings, a normal serum creatinine can represent a significant loss of GFR. Rarely, the same etiologic agent may be responsible for both the liver and kidney injury. This occurs with certain infections (e.g., leptospirosis, hantavirus) and nephrotoxic agents (Table 70.3). Cholemic nephropathy, AKI secondary to bile cast mediated tubular injury, has been described in obstructive jaundice.

AKI in Heart Failure (Cardiorenal Syndrome)

The development of AKI in patients with decompensated heart failure is common and is associated with poor prognosis. Reduced kidney perfusion secondary to decreased cardiac output has long been considered the primary cause; however, there are important contributions from right ventricular dysfunction leading to kidney venous hypertension and from activation of renin angiotensin and sympathetic nervous systems.[38]

AKI in the Cancer Patient

Patients with cancer are prone to AKI due to the underlying malignancy and its treatment (see Chapter 67). In a large European population, the incidence of AKI was 18% in the year following cancer diagnosis, and AKI occurs commonly in critically ill cancer patients.[39] Prerenal azotemia is common in cancer patients due to the high frequency of vomiting and diarrhea, and urinary tract obstruction must always be ruled out (Table 70.4). More specific causes of intrarenal AKI are noted in the following sections.

Tumor Lysis Syndrome

Necrosis of tumor cells may release large amounts of nephrotoxic intracellular contents (uric acid, phosphate, xanthine) into the circulation. This usually occurs after treatment of lymphoma (particularly Burkitt lymphoma) and leukemia but may occur with solid tumors. Rarely, a spontaneous form of tumor lysis syndrome occurs in patients with rapidly growing tumors that have outstripped their blood supply. AKI occurs when crystals of uric acid, calcium phosphate, and xanthine precipitate in the renal tubules causing obstruction and inflammation. Hyperuricemia may also contribute to AKI by crystal-independent mechanisms including renal vasoconstriction and oxidative injury.[40] Risk factors for tumor lysis syndrome include preexisting CKD, bulky and rapidly proliferating tumors, volume depletion, and acidic urine. The AKI is typically oligoanuric, and the condition should be suspected in patients with high lactate dehydrogenase levels suggestive of massive cell lysis. Markedly elevated phosphate and urate levels may be found. Hyperkalemia may be prominent and life-threatening. Prevention and therapy of tumor lysis syndrome are discussed in Chapter 67.

TABLE 70.3 Causes of AKI and Liver Disease

Prerenal azotemia	Diuretic use, gastrointestinal loss, large-volume paracentesis, hypoalbuminemia
Hepatorenal syndrome	
Acute tubular necrosis	Hyperbilirubinemia (bile cast nephropathy), sepsis, GI hemorrhage
Drugs	Acetaminophen (paracetamol), NSAIDs, tetracycline, rifampin, isoniazid, anesthetic agents, sulfonamides, anticonvulsants (DRESS syndrome), allopurinol, methotrexate, aspirin (Reye syndrome)
Infections	Hepatitis C and cryoglobulinemia, hepatitis B and polyarteritis nodosa, leptospirosis, hantavirus, Epstein-Barr virus, gram-negative sepsis, spontaneous bacterial peritonitis
Other	Papillary necrosis, inhalation of chlorinated hydrocarbons, mushroom poisoning *(Amanita phalloides),* carbon tetrachloride

AKI, Acute kidney injury; *DRESS,* drug reaction with eosinophilia and systemic symptoms; *GI,* gastrointestinal; *NSAID,* nonsteroidal antiinflammatory drug.

Hypercalcemia

Hypercalcemia is common in advanced cancer and may result from lytic bone metastases, overproduction of 1,25-dihydoxyvitamin D (predominantly lymphomas), or production of parathyroid hormone–related peptide. Hypercalcemia causes nausea, vomiting, and polyuria, and AKI in this setting is often driven by volume depletion. Additional mechanisms for hypercalcemia-induced AKI include direct intrarenal vasoconstriction and intratubular obstruction.

Chemotherapeutic Agents

Cisplatin is commonly associated with nonoliguric AKI, and nephroxicity is the most common dose-limiting side effect of this drug.[41] Cisplatin accumulates in mitochondria inhibiting oxidative phosphorylation, resulting in excessive reactive oxygen formation and impairment in ATP generation, leading to cell death. Tubular injury affects both the proximal and distal nephrons and clinically may be associated with magnesium wasting, impaired urinary concentration, and rarely salt wasting with volume depletion. Prophylaxis against nephrotoxicity includes volume loading and reducing the dose when possible. Although kidney impairment may persist after treatment, progressive decline in GFR is unusual. When combined with bleomycin and vinca alkaloids, cisplatin is also associated with thrombotic microangiopathy. The alternative agent carboplatin appears to be less nephrotoxic.

Ifosfamide is a cyclophosphamide analog with a nephrotoxic metabolite, chloroacetaldehyde. AKI is usually mild, although

TABLE 70.4 Causes of AKI in Patients With Cancer

Prerenal	Nausea and vomiting, hypercalcemia, cardiomyopathy secondary to chemotherapy
Vascular	Thrombotic microangiopathy (adenocarcinoma of stomach; cancer of the breast, prostate, lung, or pancreas; radiation nephropathy), renal vein thrombosis secondary to hypercoagulability, DIC (acute promyelocytic leukemia)
Glomerular	Rapidly progressive glomerulonephritis
Acute tubular necrosis	Sepsis and antibiotic nephrotoxicity
Malignant infiltration	Lymphoma, acute lymphoblastic leukemia
Intraluminal obstruction	Tumor lysis syndrome, myeloma cast nephropathy
Postrenal obstruction	Transitional cell carcinoma ureters and bladder, prostatic obstruction, extrinsic ureteral compression (tumor, nodes, retroperitoneal fibrosis)
Chemotherapeutic Agents	
Tubular toxicity	Cisplatin, ifosfamide, plicamycin (mithramycin), imatinib, pemetrexed, pentostatin, high-dose bisphosphonates
Thrombotic microangiopathy	Mitomycin C, gemcitabine, cisplatin, CNIs, anti-VEGF therapies
Other mechanisms	Capillary leak syndrome (IL-2 therapy), AIN (interferon α, check point inhibitors, lenalidomide), intraluminal obstruction (methotrexate)

AIN, Acute interstitial nephritis; *AKI,* acute kidney injury; *CNI,* calcineurin inhibitor; *DIC,* disseminated intravascular coagulation; *IL-2,* interleukin-2; *VEGF,* vascular endothelial growth factor.

proximal tubular dysfunction (Fanconi syndrome) and hypokalemia may be prominent and sometimes persistent.

High-dose methotrexate and its metabolites can precipitate within the tubular lumen and cause AKI. Risk factors for drug crystallization include acid urine, volume depletion, impaired GFR, and concurrent nephrotoxin exposure. Aggressive hydration and urinary alkalinization (pH >7) can reduce the risk for nephrotoxicity.

Gemcitabine is a nucleoside analog that causes AKI primarily by damaging the renal endothelium. The most common manifestation is a thrombotic microangiopathy that develops 1 to 2 months following drug completion. AKI is often accompanied by new onset hypertension or worsening of existing hypertension. Laboratory evidence of systemic hemolysis is common. AKI due to thrombotic microangiopathy may also be caused by mitomycin and by drugs that inhibit vascular endothelial growth factor pathways (e.g., bevacizumab, tyrosine kinase inhibitors).[42]

Immune checkpoint inhibitors (e.g., ipilimumab, nivolumab, and pembrolizumab) all increase the risk of AKI. Several case series have identified AIN as the predominant mechanism of AKI, although TMA and immune complex–mediated glomerulonephritis have also been observed. Additional chemotherapeutic agents that may cause AKI are listed in Table 70.4.

SELF-ASSESSMENT QUESTIONS

1. Acute tubular injury predominantly affects which segments of the nephron?
   A. Proximal convoluted tubule and loop of Henle
   B. Proximal tubule and distal convoluted tubule
   C. Loop of Henle and distal convoluted tubule
   D. Distal convoluted tubule and collecting ducts
   E. All tubular segments of the nephron equally
2. What is the most likely mechanism by which nonsteroidal antiinflammatory drugs cause AKI in a patient with systolic heart failure?
   A. Impaired efferent arteriole vasodilation
   B. Impaired efferent arteriole vasoconstriction
   C. Impaired tubuloglomerular feedback
   D. Impaired afferent arteriole vasodilation
   E. Impaired afferent arteriole vasoconstriction
3. Which of the following nephrotoxic agents causes AKI predominantly by the formation of intratubular crystals?
   A. Cisplatin
   B. Methotrexate
   C. Gentamicin
   D. Amphotericin
   E. Tacrolimus
4. How does tubular injury (acute tubular necrosis) cause a drop in glomerular filtration rate?
   A. Efferent arteriole vasoconstriction and increased hydraulic pressure in Bowman's space
   B. Afferent arteriole vasoconstriction and increased hydraulic pressure in Bowman's space
   C. Efferent arteriole vasoconstriction and decreased hydraulic pressure in Bowman's space
   D. Afferent arteriole vasoconstriction and decreased hydraulic pressure in Bowman's space
   E. Efferent arteriole vasodilation and unchanged hydraulic pressure in Bowman's space

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